PIPELINER

EARTH OBSERVATION PIPELINE

ML4EO · RESEARCH PLATFORM

Reproducible EO pipelines for machine learning

Fast acquisition, analysis and management of Earth-observation data.

pipeliner.ml4eo.world

Scan to explore the **live platform**

Everything is available – yet it doesn't land easily

We have all the tools for every single task. Using it is another matter.

Coding proficiency requirements

Google Earth Engine (GEE) and other primary EO data sources require coding at almost every step, limiting collaboration with the non-technical decision-makers who ultimately act on the results.

No home for deep models

Although EO is moving toward foundation models (Prithvi, Clay, SatMAE), mainstream deployment platforms like GEE Apps offer no ML runtime to host their inference online.

Hard to reproduce

Consequently, most EO research models remain confined to GitHub repositories, their real-world value left untapped, leaving a gap between published methods and operational tools that non-technical users can actually use.

Many things, many tools

Pipeliner: From raw data to evaluated datasets

One platform, every step — from raw data to shared, evaluated datasets.

Data Acquisition and Process

We employed an automated pipeline to handle the boiler-plate code for large-scale data acquisition and time-series data collection. Enabling data acquisition to be code free and be performed by anyone.

Analysis and Inference Online

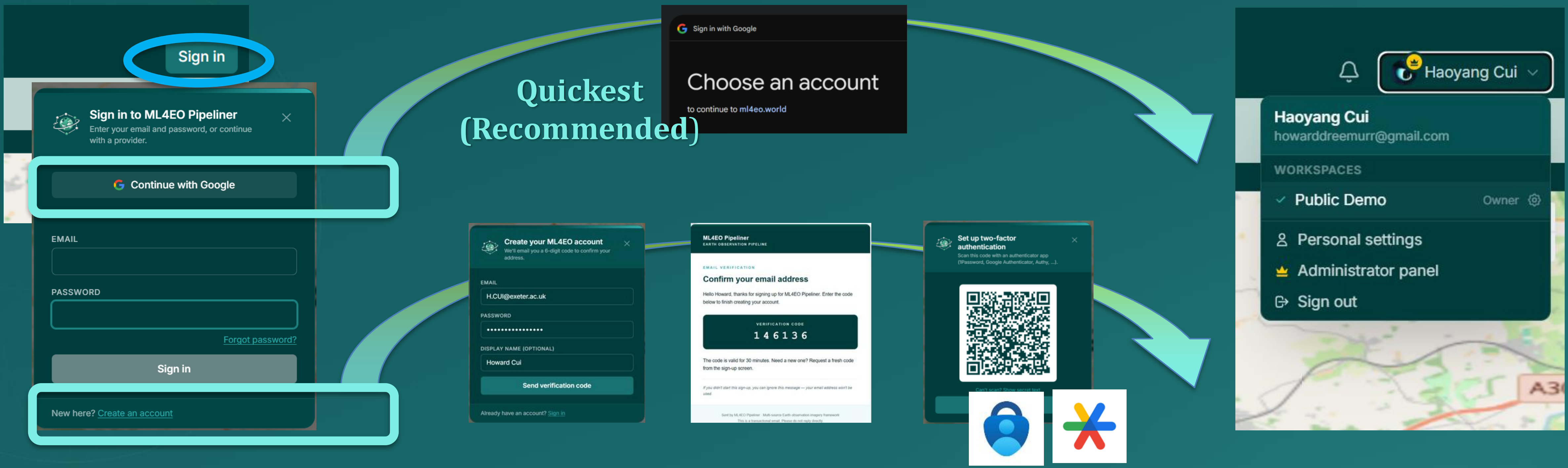
We provide a deployable runtime that hosts multiple models for code-free online inference, generating results on demand and making advanced EO models usable by anyone — not only those who can run them locally.

Collaborate and Share

Workspace allows multiple user operate on the same STAC isomorphic data collection. From quickly validate the idea to building and manage the mass dataset assets together.

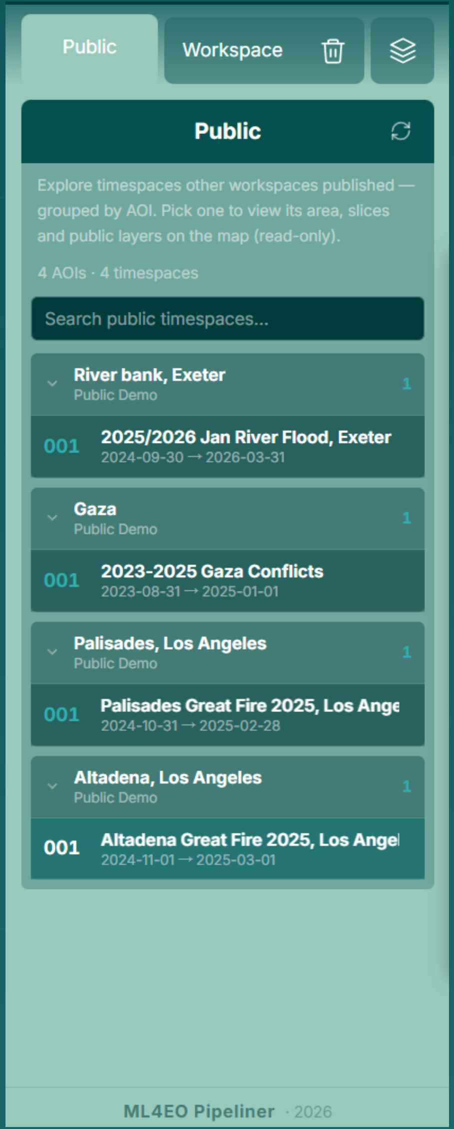
Access and Registration

Register with the platform gives you **shareable workspace** and ability to **fetch, process and manage your datasets**



Or you may use the Public DEMO without registration

Basic Control: Public or Workspace



Workspace

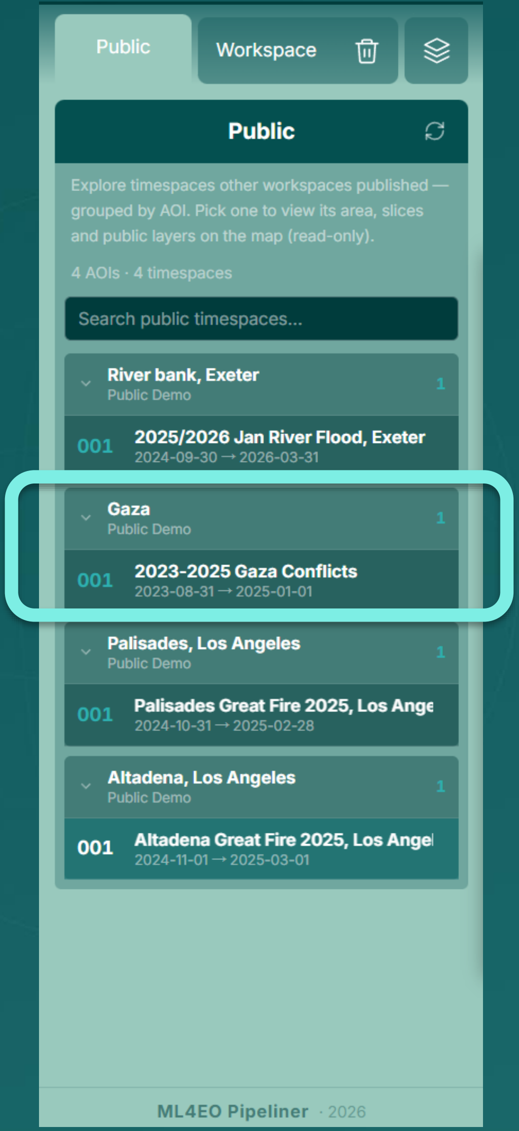
Once logged-in, you may select **Guided** or **Pro** mode

Guided mode allows you to run pre-defined fast workflow for specific task; **Pro** mode gives fine-grained operate ability, Guided results also in the **Timespace**

Public

The **Public** contains results in Public Demo and **Timespaces** that other users published

Practice I: Operate on Gaza Conflict Demo



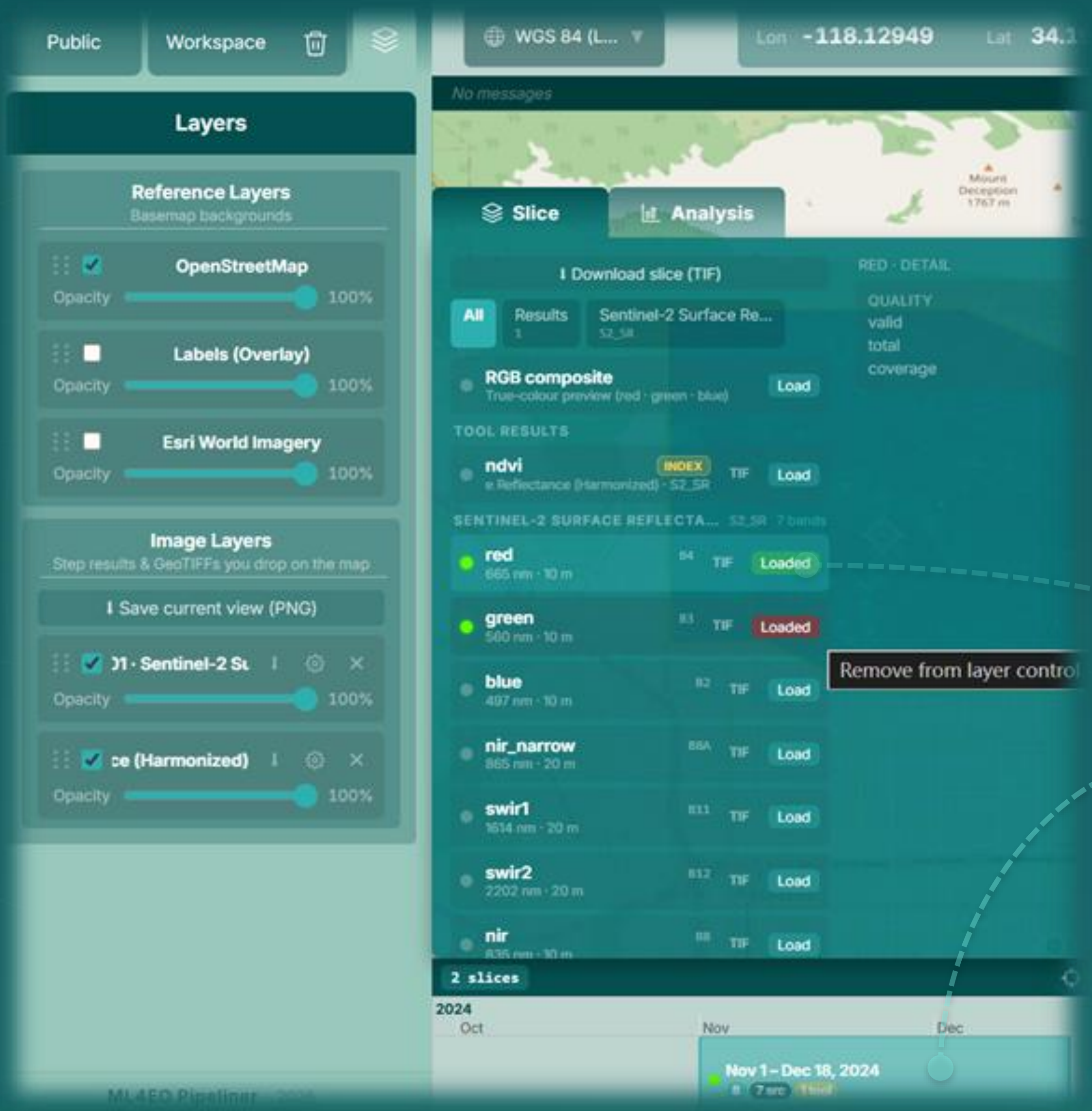
Background

The Gaza conflict escalated in October 2023 and has continued for over a year, through to the ceasefire that took effect in January 2025. Over this period, the Gaza Strip experienced widespread destruction of agricultural land and built-up areas, with satellite analyses reporting damage to the large majority of structures across the territory.

Our demo draws on time-series satellite imagery spanning September 2023 to January 2025 and tracks changes in NDVI and land cover across this window.

We hope to demonstrate the **visualization and analysis tools** with this case.

Basic Control: Layers



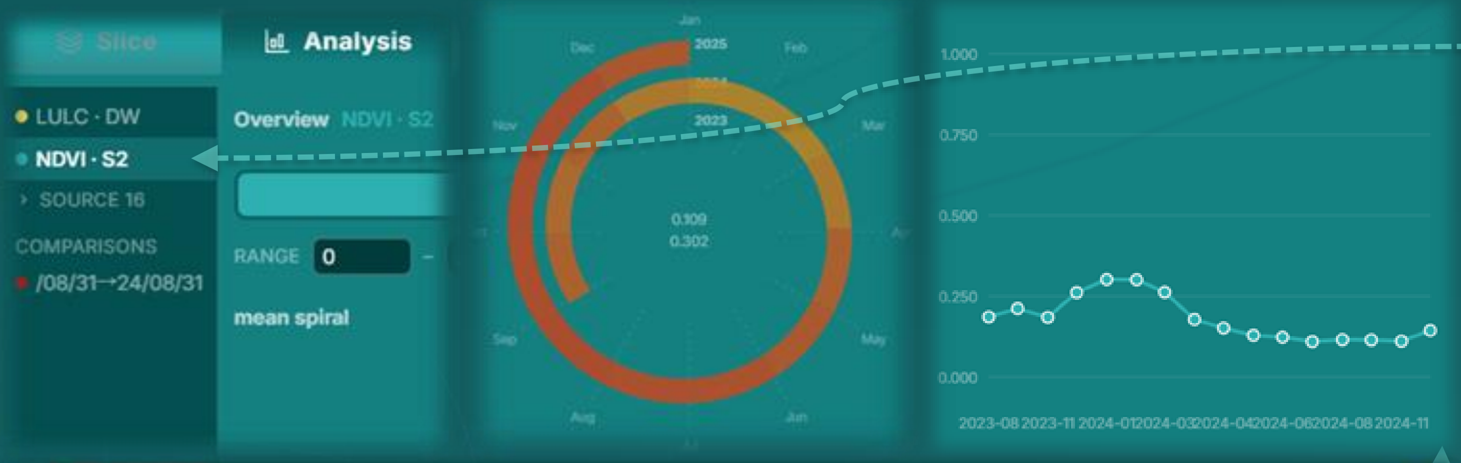
You may access the Layer Control to adjust reference/image layer opacity and rendering order

Once the **Slice** selected, you may access the **Slice Panel**
And the image layer (Which need **Timespace** Selected)

Can be temporarily appeared on map by just click the interested **Layer Tab**

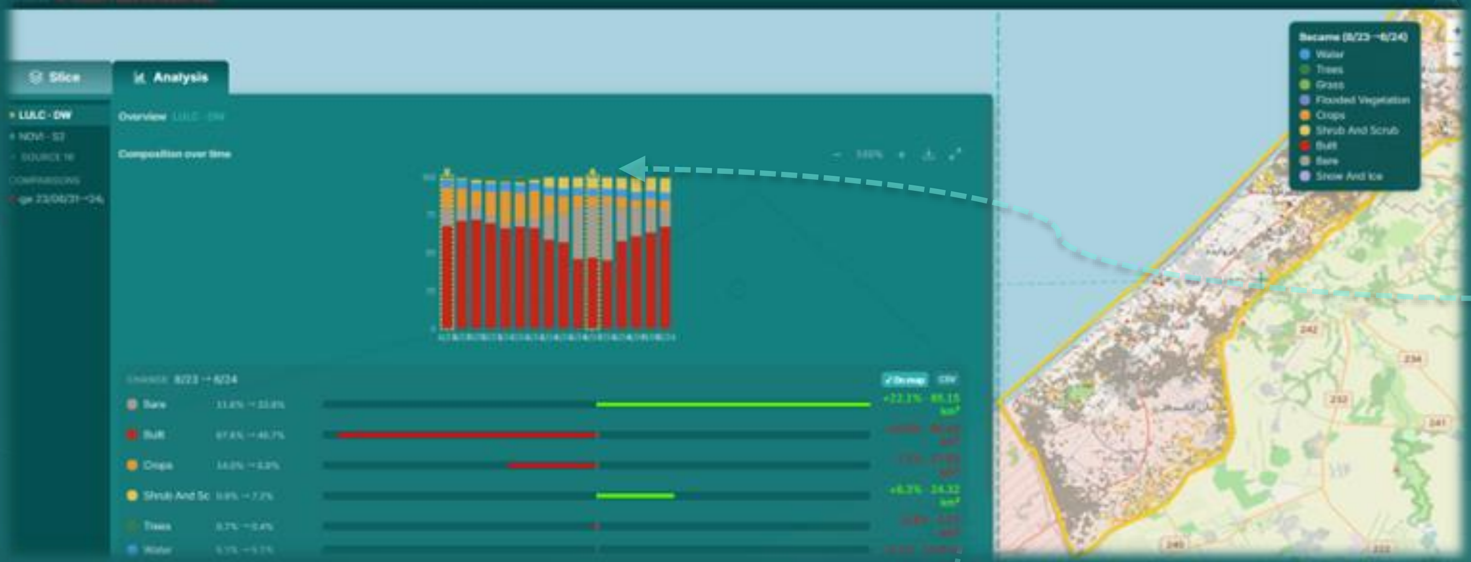
Or click "**Load**" to load into the **Image Layers Control** for adjust and visual compare later

Basic Control: Analysis



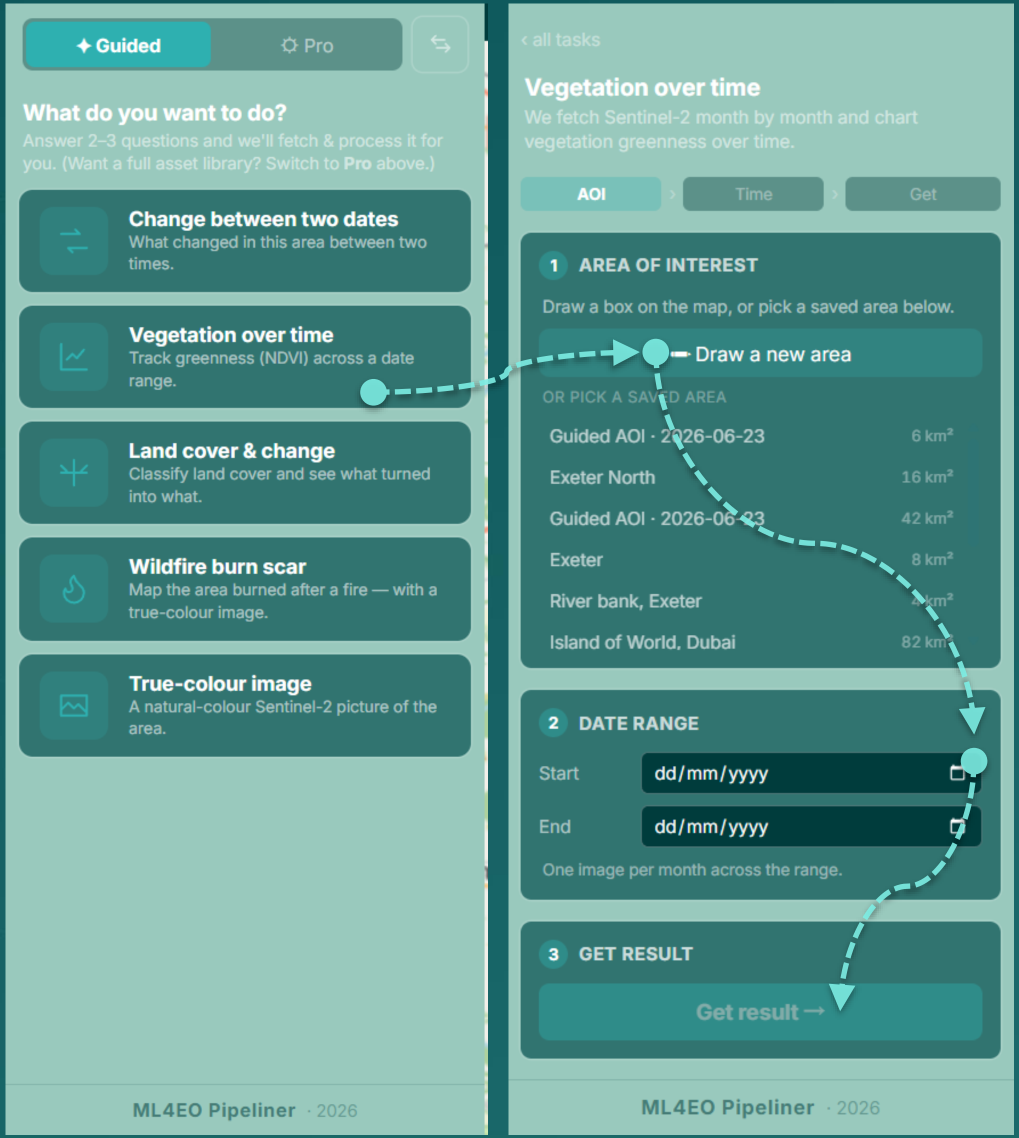
You may access the Analysis Panel if there are result layer which support time sequential comparison

Different results may use the different visualization tool to generate graphs that use for intuitive trending analysis



Comparison Mode: Click two data point on the graph will perform a First – Last comparison operation that shows the difference on both statistically and visually

Guided Mode: Quick Task Workflow



Guided mode pipeline allows you to run pre-defined fast workflow for specific task

Simply select a workflow and proceed the *AOI, TIME, GET* steps in a row

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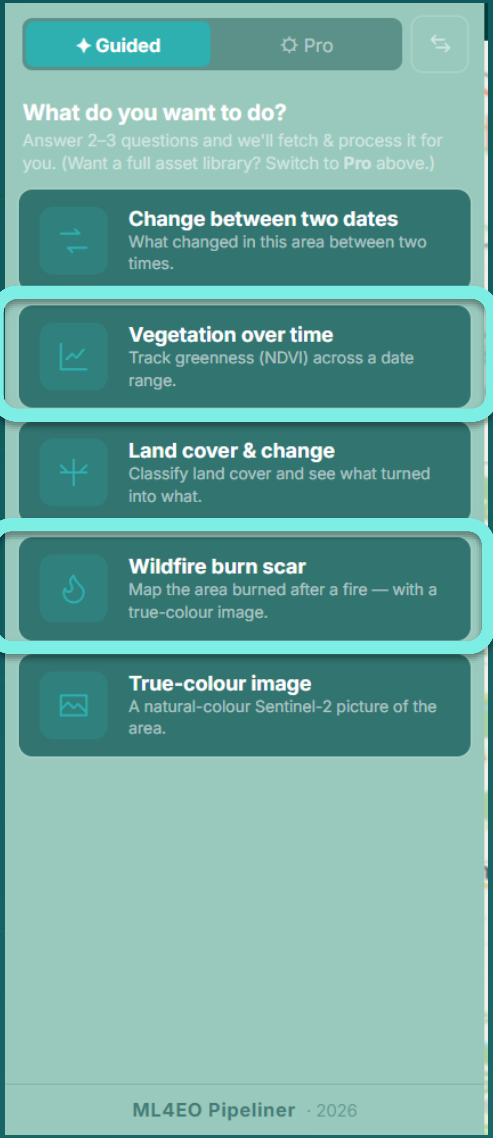
Practice II: Eaton Fire In Los Angeles 2025



Background

The Los Angeles wildfires escalated on January 7, 2025, when the Palisades and Eaton fires erupted within hours of each other and, driven by 80–100 mph Santa Ana winds, tore across opposite ends of Los Angeles County before both were contained on January 31. Over this period, the two fires burned more than 38,000 acres and destroyed over 16,000 structures across Southern California communities, leveling much of Pacific Palisades and Altadena.

We will use the ***Vegetation Over Time*** workflow to quickly fetch the Normalized Difference Vegetation Index (NDVI) to examine the vegetation change trend, the possibly affected area, and the best evaluation time window for the ***Wildfire Burn Scar*** model.

We will then apply the ***Wildfire Burn Scar model*** within that time window to assess its performance and identify its limitations.

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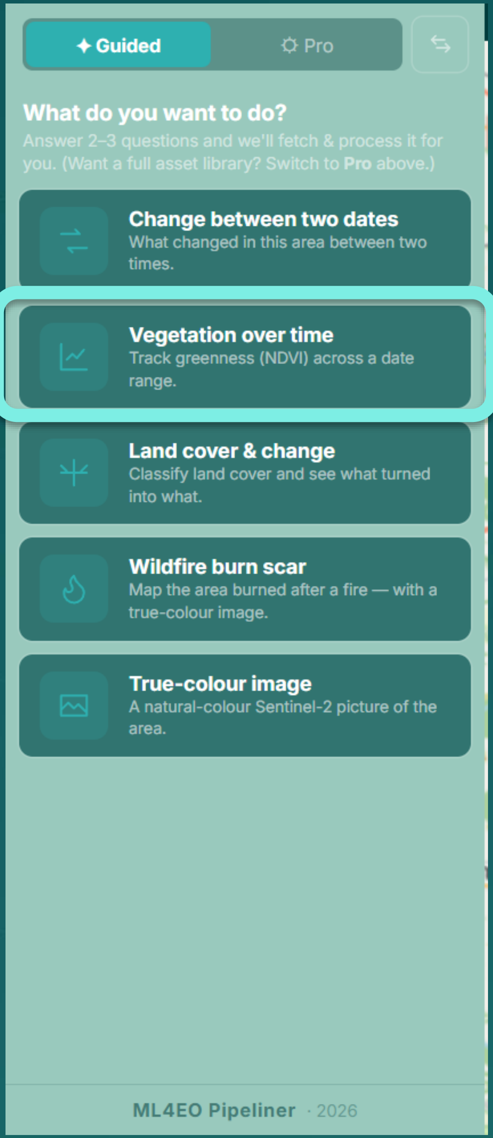
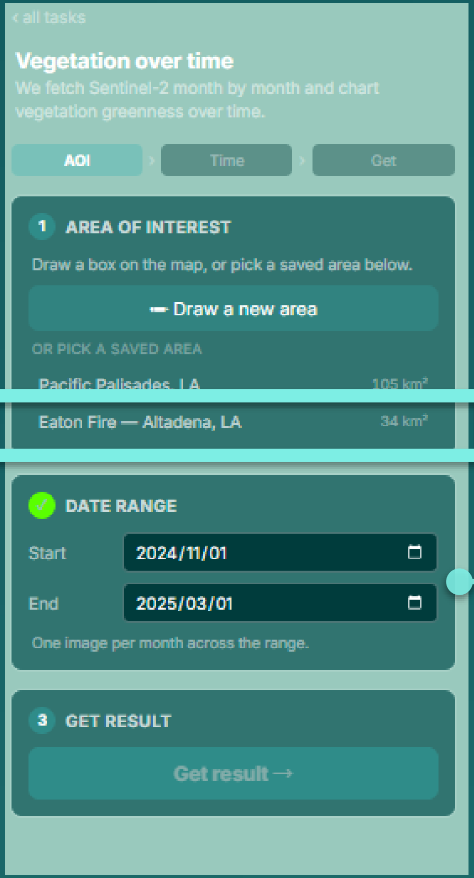
Practice II: Eaton Fire In Los Angeles 2025

Select the **Vegetation Over Time** workflow, then provide the pre-defined AOI and a date range spanning before and after the event.





AOI: Eaton Fire – Altadena, LA

Date Range: 2024/11/01 – 2025/03/01

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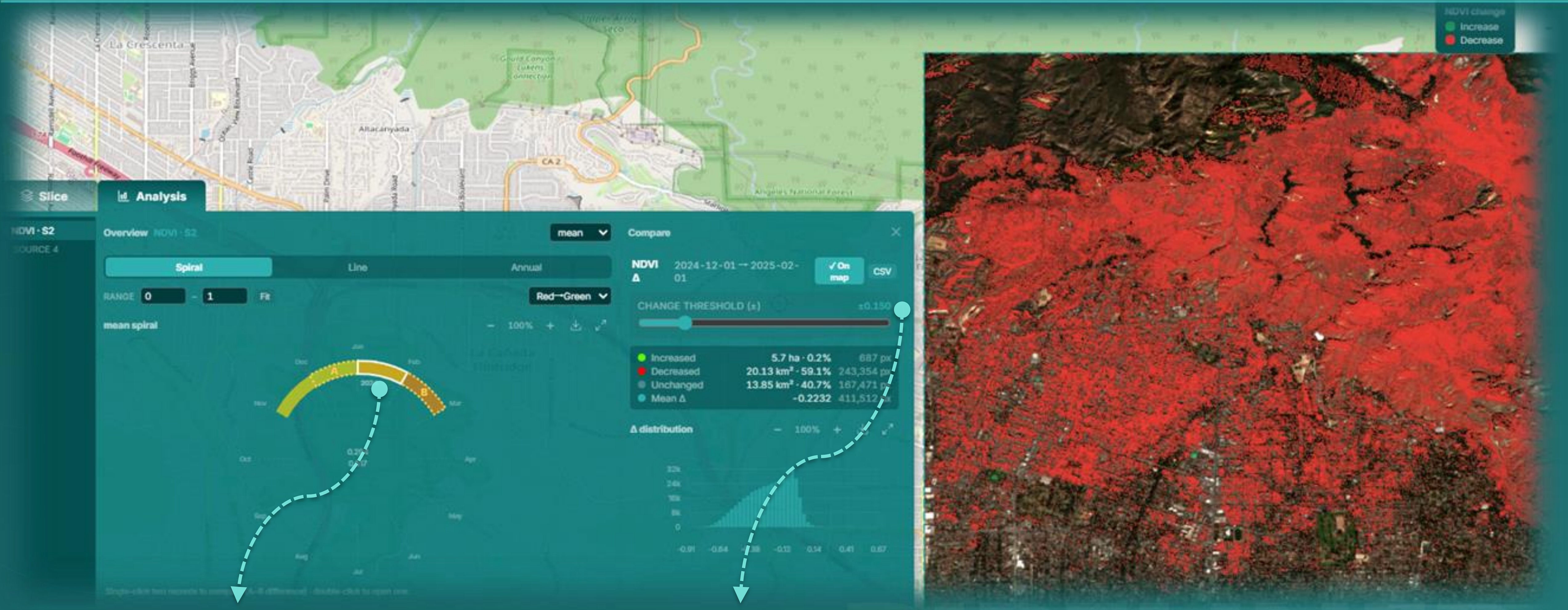
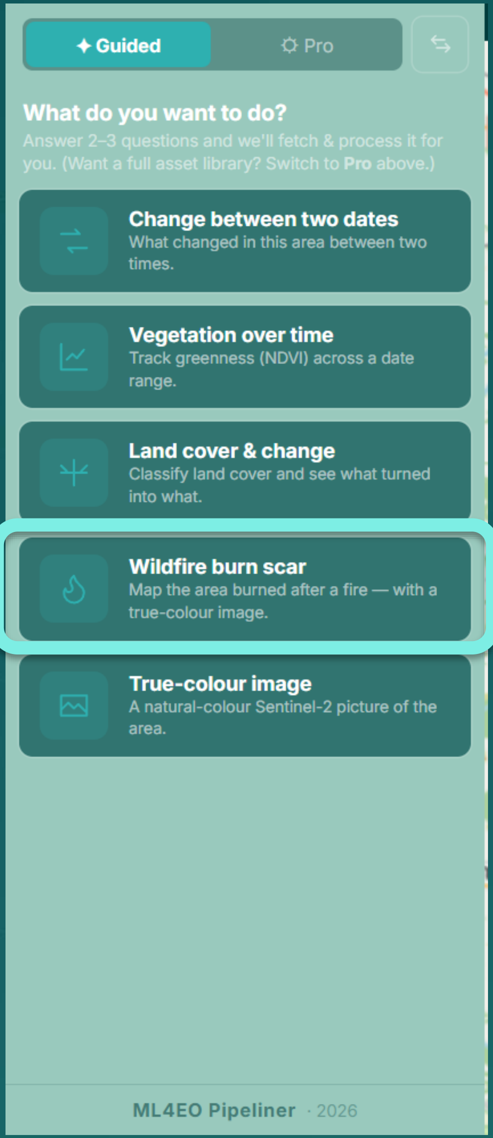
Practice II: Eaton Fire In Los Angeles 2025



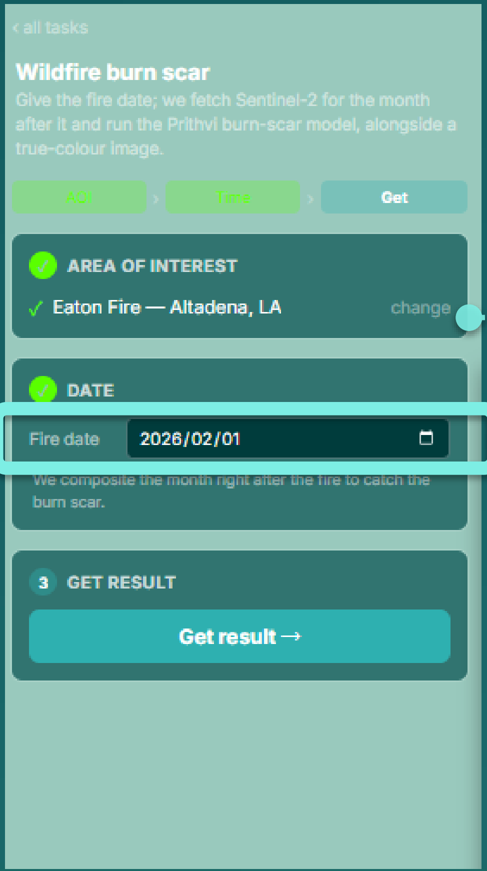
Compare: You may compare change of each month by select A-B from left to right to see the abnormal change

Change Threshold : The value is auto fitted with the data range, but here the optimal value should be around 0.15

Practice II: Eaton Fire In Los Angeles 2025



Select the **Wildfire burn scar** workflow, then provide the pre-defined AOI and a best date after we observe the abnormal.



AOI: Eaton Fire – Altadena, LA

Date: 2026/02/01

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GUILDLINE: SECTION II

Practice II: Eaton Fire In Los Angeles 2025

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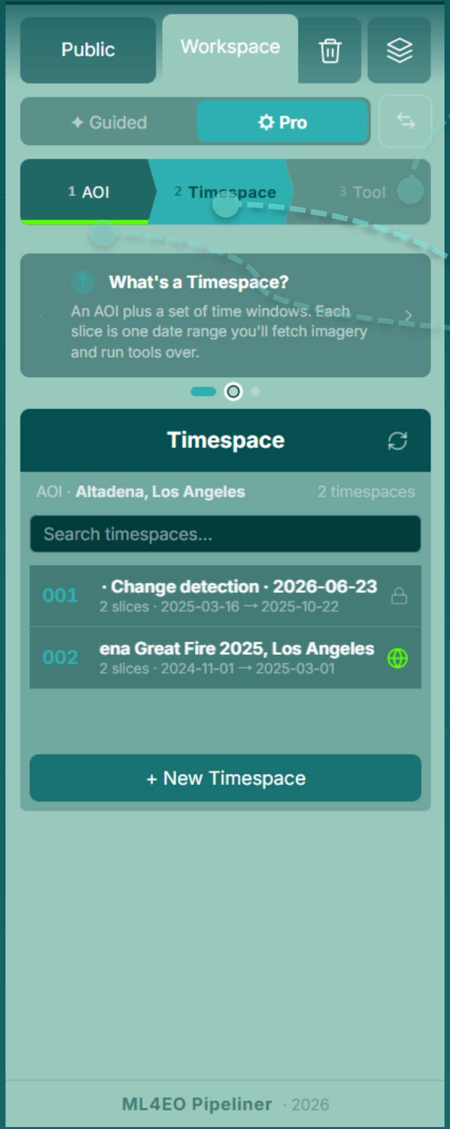
Practice III: Create your own STAC dataset

Any analysis — **ML** in particular — depends on fast access to data and a well-structured **asset catalog**.

With the **Pro mode** pipeline, we acquire the required imagery, organize it into a standards-compliant STAC catalog, and export it for downstream analysis and model development.

Pro Mode: Fine-grained Operations



Pro mode pipeline consists with 3 steps: AOI, Timespace, and Tool Run

AOI: Area of Interests (AOI) Selection

Timespace: Time slices define with the AOI *(STAC's Collection)*

Tool: Automatic imagery source fetch and evaluation

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Pro Mode: AOI Selection

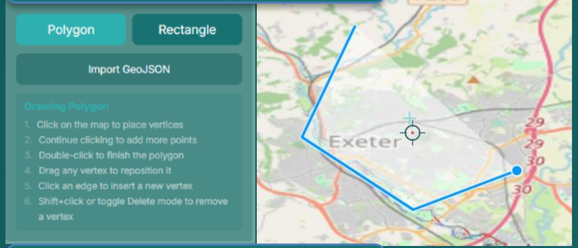
AOI Selection require to select one of the three Drawing Tool to

① define your AOI

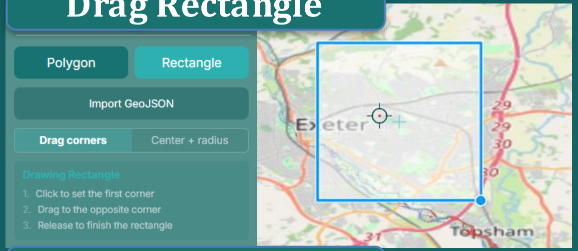
② and register it with a name

③ then proceed to Timespace

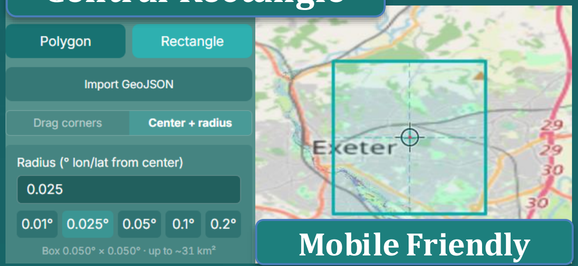
Free Polygon

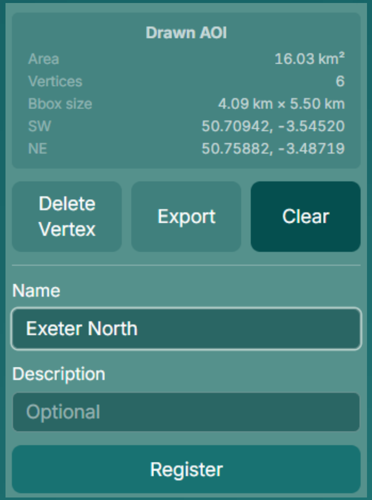


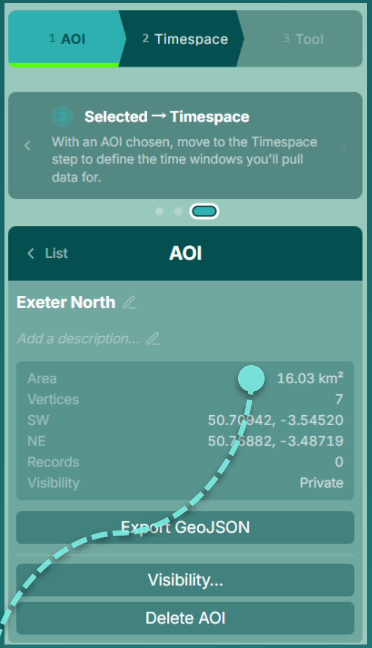
Drag Rectangle



Central Rectangle







Mobile Friendly

Note: Every Timespace have 150 km² default size limitation for each AOI, you may change it in the workspace setting up to 2000 km² with cautious

Pro Mode: Timespace Define

Timespace will create an instance with selected AOI and Slices by

① creating a Timespace with name

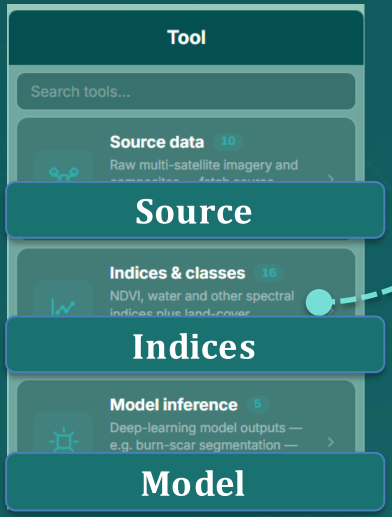
② and create time Slices and stage

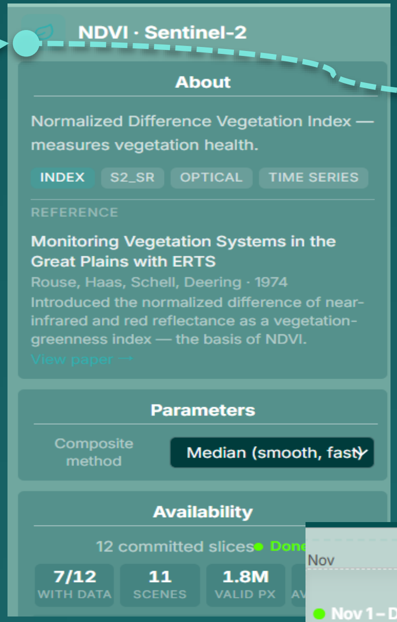
③ then commit and time for Tool run

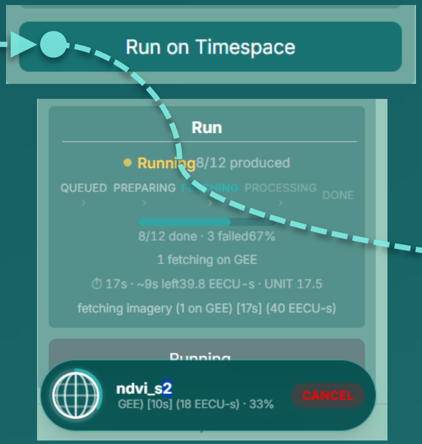
Note: You may also ***publish*** the selected Timespace so other users may access it in Public, or ***export*** the whole Timespace as a Collection ZIP in STAC

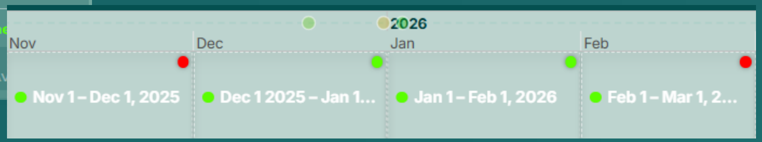
Pro Mode: Tool Run

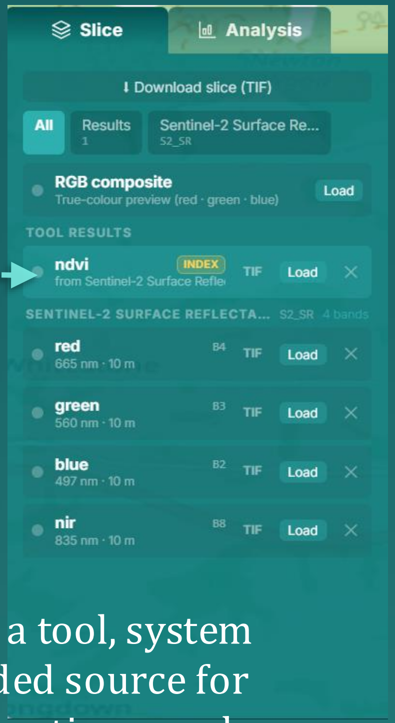
Tool Run will auto fetch the source and operate the evaluation to generate relative Imagery Layers

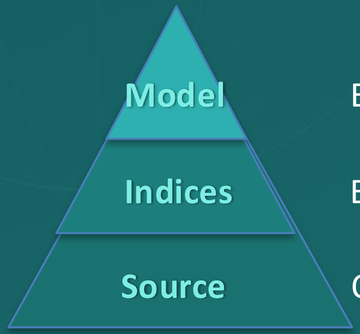












Evaluated Imagery data with ML model from **Source**

Evaluated Imagery data with numerical computation from **Source**

Original Imagery data from all range of EO resource provider

Availability: Once you selected a tool, system will check availability of depended source for each slice to see if there is observations, each dots represents an available data point, and Slice marked red means no data at that period

Vision and Further Planning

This is just the beginning, we aim for better functionality and community support

Better Batch Operation

Queue many AOIs and time-slices in a single submission and run them as one batch job — no more launching tasks one by one.

Better Data Availability Observation

A dedicated tool to scan image coverage and cloud-free availability across very large AOIs, so you can pick the optimal time window before a large-scale mapping run.

Better Time Resolution

Finer temporal granularity than the current slice-level aggregation, with smarter automatic filtering of scenes by quality and cloud cover.

Our Platform

We'll keep ML4EO as the flagship instance, onboarding more models from teams who want their work showcased and used.

Your Own Instance

A deployment package for organizations to host on their own infrastructure. With your own local data storage and better internal collaboration, fully self-contained.

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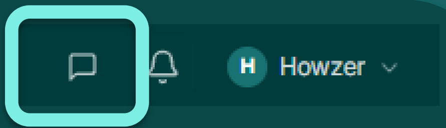
Help Us to Shape What's the Next

A platform is nothing without the people who **use** it.

We want Pipeliner to genuinely help the people who need it. Every piece of **feedback** and every **suggestion** is the greatest support you can give — and the most valuable contribution to the platform.

Feedback



Share feature requests and comments any time, right where you work.

→ Use the **Feedback** button on the platform

Contact

Have bigger or more specialized requirements? Reach out directly and we'll work with you on what you need.

→ Email **C.Luo@exeter.ac.uk**